

Homework 1: Deep Generative Models (ESE 6450-001)

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Due Date: 09/29/2026, 12:00 PM (Noon) Eastern Time

Quiz Date: 09/29/2026, 1:45 PM Eastern Time

Instructions.

- You must submit your solutions by the stated due date and time to receive credit. However, the submitted assignment itself will not be graded directly. Instead, your grade will be determined by a quiz designed to assess your understanding of the assignment solutions and the related course material. The quiz will be closed-book, with no access to computers, phones, or the internet. If needed, a reference sheet will be provided with the quiz.
- The purpose of the assignment is to help you learn the course material. Ideally, you should complete it independently, without assistance from other people or AI tools. You may discuss the material with other students in the class and use AI tools to support your learning, but you may not seek or copy complete solutions. Effective use of AI typically involves questioning its responses, requesting clarification or additional information, identifying errors in its reasoning, and iterating until you understand the solution. Copying solutions from another person or an AI tool is not permitted. You must write the solutions yourself and be able to explain them.
- Please submit one .pdf file containing your responses to the analytical and programming questions, as well as one .zip file containing the source code needed to reproduce the results reported in the programming questions. For analytical problems, typed solutions are preferred, but clearly handwritten solutions are also acceptable. For programming problems, include all necessary figures and explanations in the .pdf file.

Analytical Questions

1. **Marginal and Conditional Distributions of a Multivariate Gaussian.** Let $\mathbf{x}$ be a D -dimensional multivariate Gaussian random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Let us split $\mathbf{x}$ into two components:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix} \in \mathbb{R}^D, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \in \mathbb{S}_{++}^D, \quad \boldsymbol{\Sigma}^{-1} = \mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix} \in \mathbb{S}_{++}^D. \quad (1)$$

- (a) Let $\mathbf{y} = \mathbf{W}\mathbf{x}$, where $\mathbf{W} \in \mathbb{R}^{d \times D}$. Show that $\mathbf{y} \sim \mathcal{N}(\mathbf{W}\boldsymbol{\mu}, \mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top)$ and that $\mathbf{x}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$ for $i = 1, 2$.
- (b) Using the identity $\mathbf{S}\boldsymbol{\Sigma} = \mathbf{I}_D$, prove that

$$\mathbf{S}_{11}^{-1} \mathbf{S}_{12} = -\boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}, \quad (2)$$

$$\mathbf{S}_{11}^{-1} = \boldsymbol{\Sigma}_{11}^{-1} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}. \quad (3)$$

- (c) Use 1b and the relationship $p(\mathbf{x}_1 | \mathbf{x}_2) = p(\mathbf{x}_1, \mathbf{x}_2) / p(\mathbf{x}_2)$ to show that $\mathbf{x}_1 | \mathbf{x}_2 \sim \mathcal{N}(\boldsymbol{\mu}_{1|2}, \boldsymbol{\Sigma}_{1|2})$, where:

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{x}_2 - \boldsymbol{\mu}_2), \quad (4)$$

$$\boldsymbol{\Sigma}_{1|2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}. \quad (5)$$

2. **Multivariate Linear Gaussian Model.** Let $\mathbf{x} \in \mathbb{R}^D$ and $\mathbf{y} \in \mathbb{R}^d$ be random variables. Consider the model

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b} + \boldsymbol{\epsilon}, \quad (6)$$

where $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ are independent, $\mathbf{W} \in \mathbb{R}^{d \times D}$ and $\mathbf{b} \in \mathbb{R}^d$.

- (a) Write down $p(\mathbf{x})$, $p(\mathbf{y} | \mathbf{x})$, and $p(\mathbf{x}, \mathbf{y})$.
- (b) Compute the entropy $h(\mathbf{x})$, conditional entropy $h(\mathbf{y} | \mathbf{x})$, and mutual information $I(\mathbf{x}; \mathbf{y})$.

- (c) Assume $d = 1$ and $\mathbf{b} = 0$, so that $y = \mathbf{w}^\top \mathbf{x} + \epsilon$. Find a unit-norm vector $\mathbf{w}$ that maximizes $I(\mathbf{x}; y)$.
- (d) Given N i.i.d. observations $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$, write down their joint log-likelihood and find the maximum-likelihood estimates of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, $\mathbf{W}$, $\mathbf{b}$, and σ^2 .
3. **Expectation Maximization for PPCA.** In class, we showed that the ML parameters $\boldsymbol{\theta} = (\mathbf{W}, \mathbf{b}, \sigma)$ of the PPCA model $\mathbf{x} = \mathbf{W}\mathbf{z} + \mathbf{b} + \epsilon$, where $\mathbf{b} \in \mathbb{R}^D$, $\mathbf{W} \in \mathbb{R}^{D \times d}$, $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I}_d)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_D)$, can be found in closed form, as also shown in Theorem 2.9 of the GPCA textbook. An alternative approach, which can be advantageous for large D , is to view $\mathbf{z}$ as a latent variable and use the EM algorithm to find the ML estimate. In this exercise, you will derive an EM algorithm for PPCA. In class, we derived the E-step. That is, assuming we have an estimate of the parameters $\boldsymbol{\theta}^{(k)} = (\mathbf{W}^{(k)}, \mathbf{b}^{(k)}, \sigma^{(k)})$ at iteration k , we computed the posterior distribution of the hidden variables $w_j^{(k)}(\mathbf{z}) = p_{\boldsymbol{\theta}^{(k)}}(\mathbf{z} | \mathbf{x}_j)$. Specifically, we showed that

$$w_j^{(k)}(\mathbf{z}) = \mathcal{N}\left(\langle \mathbf{z}_j \rangle^{(k)}, \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}}^{(k)}\right) \quad (7)$$

where

$$\langle \mathbf{z}_j \rangle^{(k)} := \left(\mathbf{W}^{(k)\top} \mathbf{W}^{(k)} + (\sigma^{(k)})^2 \mathbf{I}_d \right)^{-1} \mathbf{W}^{(k)\top} (\mathbf{x}_j - \mathbf{b}^{(k)}) \quad \text{and} \quad (8)$$

$$\boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}}^{(k)} := (\sigma^{(k)})^2 \left(\mathbf{W}^{(k)\top} \mathbf{W}^{(k)} + (\sigma^{(k)})^2 \mathbf{I}_d \right)^{-1}. \quad (9)$$

- (a) Let $\langle \mathbf{z}_j \mathbf{z}_j^\top \rangle^{(k)} := \boldsymbol{\Sigma}_{\mathbf{z}|\mathbf{x}}^{(k)} + \langle \mathbf{z}_j \rangle^{(k)} \langle \mathbf{z}_j \rangle^{(k)\top}$. Show that, up to terms independent of $(\mathbf{W}, \mathbf{b}, \sigma)$, the expected complete-data log-likelihood $Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(k)}) := \sum_{j=1}^N \mathbb{E}_{\mathbf{z} \sim w_j^{(k)}} \log p_{\boldsymbol{\theta}}(\mathbf{x}_j, \mathbf{z})$ can be written as

$$-\frac{DN}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^N \left(\|\mathbf{x}_j - \mathbf{b}\|^2 - 2(\mathbf{x}_j - \mathbf{b})^\top \mathbf{W} \langle \mathbf{z}_j \rangle^{(k)} + \text{trace} \left(\mathbf{W}^\top \mathbf{W} \langle \mathbf{z}_j \mathbf{z}_j^\top \rangle^{(k)} \right) \right). \quad (10)$$

Hint: You do not need to compute complicated integrals for this question. The following facts are useful

$$\langle \mathbf{z}_j \rangle^{(k)} = \int_{\mathbf{z}} w_j^{(k)}(\mathbf{z}) \mathbf{z} d\mathbf{z} \quad \text{and} \quad \langle \mathbf{z}_j \mathbf{z}_j^\top \rangle^{(k)} = \int_{\mathbf{z}} w_j^{(k)}(\mathbf{z}) \mathbf{z} \mathbf{z}^\top d\mathbf{z}. \quad (11)$$

- (b) Let $\mathbf{b}^{(k)} = \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j$. Show that the parameters $(\mathbf{W}, \mathbf{b}, \sigma)$ that maximize $Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(k)})$ are

$$\mathbf{W}^{(k+1)} = \sum_{j=1}^N \left(\mathbf{x}_j - \mathbf{b}^{(k)} \right) \langle \mathbf{z}_j \rangle^{(k)\top} \left(\sum_{j=1}^N \langle \mathbf{z}_j \mathbf{z}_j^\top \rangle^{(k)} \right)^{-1}, \quad (12)$$

$$\sigma^{(k+1)} = \left(\frac{1}{ND} \sum_{j=1}^N \left[\|\mathbf{x}_j - \mathbf{b}^{(k)}\|^2 - 2(\mathbf{x}_j - \mathbf{b}^{(k)})^\top \mathbf{W}^{(k+1)} \langle \mathbf{z}_j \rangle^{(k)} + \text{trace} \left(\mathbf{W}^{(k+1)\top} \mathbf{W}^{(k+1)} \langle \mathbf{z}_j \mathbf{z}_j^\top \rangle^{(k)} \right) \right] \right)^{1/2}. \quad (13)$$

4. **Evidence Lower Bound (ELBO) for PPCA.** Consider the PPCA model $p_{\boldsymbol{\theta}}(\mathbf{x})$ defined as $\mathbf{x} = \mathbf{W}\mathbf{z} + \mathbf{b} + \epsilon$, where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I}_d)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_D)$ are independent random variables, and $\boldsymbol{\theta} = (\mathbf{W}, \mathbf{b}, \sigma)$ are model parameters, with $\mathbf{W} \in \mathbb{R}^{D \times d}$, $\mathbf{b} \in \mathbb{R}^D$, and $\sigma > 0$. In this exercise, you will show that maximizing the ELBO with respect to a linear encoder is sufficient for the PPCA model. Specifically, let $\{\mathbf{x}_i\}_{i=1}^N$ be N i.i.d. samples from $p_{\boldsymbol{\theta}}(\mathbf{x})$, and let $q_{\boldsymbol{\psi}}(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\mathbf{A}\mathbf{x} + \mathbf{c}; \boldsymbol{\Sigma}_e)$, where $\boldsymbol{\psi} := (\mathbf{A}, \mathbf{c}, \boldsymbol{\Sigma}_e)$, $\mathbf{A} \in \mathbb{R}^{d \times D}$, $\mathbf{c} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma}_e \in \mathbb{S}_{++}^d$. For the above $p_{\boldsymbol{\theta}}$ and $q_{\boldsymbol{\psi}}$ you will show that

$$\sum_{i=1}^N \log(p_{\boldsymbol{\theta}}(\mathbf{x}_i)) = \max_{\boldsymbol{\psi}} \left\{ \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\psi}) := \sum_{i=1}^N [\mathbb{E}_{q_{\boldsymbol{\psi}}(\mathbf{z}_i | \mathbf{x}_i)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i | \mathbf{z}_i) - D_{\text{KL}}(q_{\boldsymbol{\psi}}(\mathbf{z}_i | \mathbf{x}_i) \| p(\mathbf{z}_i))] \right\}. \quad (14)$$

- (a) Recall that $p_{\theta}(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \mathbf{b}, \sigma^2 \mathbf{I})$. Show that

$$\mathbb{E}_{q_{\psi}(\mathbf{z}_i \mid \mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i \mid \mathbf{z}_i) = -\frac{D}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left[\|\mathbf{x}_i - \mathbf{W}(\mathbf{A}\mathbf{x}_i + \mathbf{c}) - \mathbf{b}\|_2^2 + \text{trace}(\mathbf{W}\Sigma_e\mathbf{W}^{\top}) \right] \quad (15)$$

and

$$D_{\text{KL}}(q_{\psi}(\mathbf{z}_i \mid \mathbf{x}_i) \parallel p(\mathbf{z}_i)) = \frac{1}{2} [\text{trace}(\Sigma_e) + \|\mathbf{A}\mathbf{x}_i + \mathbf{c}\|_2^2 - d - \log \det(\Sigma_e)]. \quad (16)$$

- (b) Fix the model parameters $\theta = (\mathbf{W}, \mathbf{b}, \sigma)$. Compute the first-order optimality conditions for maximizing the ELBO with respect to the encoder parameters $\psi = (\mathbf{A}, \mathbf{c}, \Sigma_e)$, and show that they can be written as

$$\mathbf{M}\mathbf{A}\Sigma_N = \mathbf{W}^{\top}\Sigma_N, \quad (17)$$

$$\mathbf{M}(\mathbf{A}\mu_N + \mathbf{c}) = \mathbf{W}^{\top}(\mu_N - \mathbf{b}), \quad (18)$$

$$\Sigma_e^{-1} = \frac{1}{\sigma^2} \mathbf{M}. \quad (19)$$

where $\mathbf{M} := \sigma^2 \mathbf{I}_d + \mathbf{W}^{\top}\mathbf{W}$, $\mu_N := \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i$, and $\Sigma_N := \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \mu_N)(\mathbf{x}_i - \mu_N)^{\top}$.

- (c) Assume that Σ_N is positive definite. Solve the first-order optimality conditions from part (b) and show that the optimal encoder parameters are

$$\hat{\mathbf{A}} = \mathbf{M}^{-1}\mathbf{W}^{\top}, \quad (20)$$

$$\hat{\mathbf{c}} = -\mathbf{M}^{-1}\mathbf{W}^{\top}\mathbf{b}, \quad (21)$$

$$\hat{\Sigma}_e = \sigma^2 \mathbf{M}^{-1}. \quad (22)$$

Write down the resulting distribution $q_{\hat{\psi}}(\mathbf{z} \mid \mathbf{x})$ and compare it with the exact posterior $p_{\theta}(\mathbf{z} \mid \mathbf{x})$ derived in class. What does this imply about the tightness of the ELBO and the expressive power of the linear Gaussian encoder for PPCA?

Programming Questions

1. **Three Algorithms for PPCA on MNIST and EYaleB.** In this problem, you will implement three different algorithms for PPCA which we have either learned in class or derived in previous problems: closed-form, EM, and ELBO. You will run the algorithms on the EYaleB and MNIST datasets and compare their performance. Refer to the Chapter 4.2 (Probabilistic Principal Component Analysis) of the book *Deep Generative Models*, as well as the Chapter 2.2.2 (PPCA by Maximum Likelihood) of *Generalized Principal Component Analysis*. They contain all the necessary details of how PPCA is constructed. We assume the following setup:

- We consider continuous random variables only, i.e., $\mathbf{z} \in \mathbb{R}^d$ and $\mathbf{x} \in \mathbb{R}^D$.
- The distribution of $\mathbf{z}$ is the standard Gaussian, i.e., $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$.
- The dependency between $\mathbf{z}$ and $\mathbf{x}$ is linear with a Gaussian additive noise: $\mathbf{x} = \mathbf{W}\mathbf{z} + \mathbf{b} + \epsilon$, where $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$, $\mathbf{W} \in \mathbb{R}^{D \times d}$, $\mathbf{b} \in \mathbb{R}^D$ and $\epsilon \in \mathbb{R}^D$.

Use the provided notebook `hw1-code/programming_ppca.ipynb`. Submit both the `.ipynb` notebook and responses to the questions. You may answer the questions directly in the notebook, export the notebook with the outputs of each cell as a `.pdf` file, and append it to the written response.

- (a) **Implement Closed-form PPCA.** Implement functions that, given samples $\{\mathbf{x}_j\}_{j=1}^N$ and latent dimension d , return the ML estimators $(\mathbf{W}^*, \mathbf{b}^*, \sigma^*)$ for PPCA. Your implementation should follow the closed-form solution discussed in class.
- (b) **Implement EM for PPCA.** Using your analytical derivations, implement one EM iteration that maps $(\mathbf{b}^{(k)}, \mathbf{W}^{(k)}, \sigma^{(k)})$ to $(\mathbf{b}^{(k+1)}, \mathbf{W}^{(k+1)}, \sigma^{(k+1)})$. Initialize with $\mathbf{b}^{(0)} = \mathbf{0}$, $\mathbf{W}^{(0)} = [\mathbf{I}_D]_{:,1:d}$, and $\sigma^{(0)} =$

- 0.1. At each iteration, compute and output the data log-likelihood $\sum_{j=1}^N \log p_{\theta^{(k)}}(\mathbf{x}_j)$ which will be used later for evaluation.
- (c) **Gradient ascent for the PPCA ELBO.** Implement gradient ascent on the PPCA ELBO with respect to $(\mathbf{W}, \mathbf{b}, \sigma, \mathbf{A}, \mathbf{c}, \Sigma_e)$ with Adam optimizer. Use the same initialization as above. You should have a hyperparameter for the learning rate, so that you can try multiple values later. Again, you should output the log-likelihood per iteration for evaluation.
- (d) **Evaluation on EYaleB and MNIST.** Now run all three methods on *one class* of EYaleB and MNIST with the same latent dimension d . The PyTorch dataloader for the EYaleB Face dataset and the data visualization are provided to you in the Jupyter notebook. Make one plot for each of the following metrics, where the x -axis is the iteration, the y -axis is the metric, and different methods correspond to different curves on the same plot. The closed-form method may appear as a horizontal line where applicable.
- **Log-likelihood:** $\sum_{j=1}^N \log p_{\theta^{(k)}}(\mathbf{x}_j)$; note that all three methods aim to maximize this quantity.
 - **Wall-clock time:** cumulative time from the beginning of the run to the current iteration.
 - **KL divergence:** $\text{KL}(\mathcal{N}(\mathbf{b}^*, \mathbf{W}^* \mathbf{W}^{*\top} + (\sigma^*)^2 \mathbf{I}_D) \parallel \mathcal{N}(\mathbf{b}^{(k)}, \mathbf{W}^{(k)} \mathbf{W}^{(k)\top} + (\sigma^{(k)})^2 \mathbf{I}_D))$; since both distributions are Gaussian, the KL divergence can be computed in closed form.
 - **$\mathbf{W}$ -subspace error:** principal angle distance in degrees between $\text{span}(\mathbf{W}^{(k)})$ and $\text{span}(\mathbf{W}^*)^1$.
 - **$\mathbf{b}$ -relative error:** $\|\mathbf{b}^* - \mathbf{b}^{(k)}\| / \|\mathbf{b}^*\|$.
 - **σ -relative error:** $|\sigma^* - \sigma^{(k)}| / |\sigma^*|$.

Briefly compare EM, ELBO, and closed-form in terms of efficiency, accuracy, and ease of hyperparameter choice. Which method would you use in practice and why?

- (e) **Effect of noise level.** For each method, use the estimated parameters to define a PPCA model, sample from it and visualize the samples. Vary the noise level and describe the quality of the image generation.
- (f) **Effect of latent dimension d .** Describe the quality of generation with respect to d .
- (g) **Effect of sample size N .** Keep d fixed. Compare the performance when using many samples ($N \gg d$), a moderate number of samples ($N = d$), and a small number of samples $N < d$.
- (h) **Effect of multiple classes.** Suppose now we move into a two/three-class regime, by adding classes to the CLASSES list. What do you observe about the sample mean? Given the data samples we have, do the images generated make sense?

¹Form orthonormal bases $U^{(k)}, U^*$ of the two column spaces; if Σ are the singular values of $(U^{(k)})^\top U^*$, principal angles are $\arccos(\Sigma_i)$; report the largest angle. For reference, see: Knyazev, A. V., & Argentati, M. E. (2002). Principal angles between subspaces in an A-based scalar product: algorithms and perturbation estimates. SIAM Journal on Scientific Computing, 23(6), 2008-2040. For implementation, you may use `scipy.linalg.subspace_angles`